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Screening for Colorectal Cancer and Evolving Issues for Physicians and Patients
A Review

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IMPORTANCE Colorectal cancer (CRC) is the second-leading cause of cancer death in the United States. Screening can reduce CRC mortality and incidence, and numerous screening options, although available, complicate informed decision making. This review provides evidence-based tools for primary care physicians to identify patients with higher-than-average-risk and engage patients in informed decision making about CRC screening options.

OBSERVATIONS Recently, the US Preventive Services Task Force recommended any of 8 CRC screening approaches for average-risk individuals, beginning at age 50 years. Only 2 methods have been shown in randomized clinical trials to reduce mortality: fecal occult blood testing and flexible sigmoidoscopy. Of the 8 programs, screenings using the fecal immunochemical test annually and colonoscopy every 10 years are now the most commonly used tests in the United States and among the most effective in reducing CRC mortality as determined by decision models. With the exception of primary screening using colonoscopy, all of the other screening approaches have multiple steps. Adherence to each phase of a multistep program is critical to achieving maximal effectiveness of the screening program. It is likely that each of the recommended programs can reduce CRC mortality, but other key outcomes may differ such as lifetime burden of colonoscopy, complications, patient acceptance, and cost. Decisions about the timing of screening cessation should be individualized.

CONCLUSIONS AND RELEVANCE CRC screening is effective if patients adhere to the steps in each screening program. There is no evidence that one program is superior to another. Informed decision-making tools are provided to assist patients and clinicians with the goal of improving adherence to effective screening.

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Supplemental content

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Colon cancer is the second leading cause of cancer death in developed countries. In the United States, 135 000 new cases and 49 000 colorectal cancer (CRC)-related deaths are estimated for 2016.[1] CRC is one of the most preventable cancers when screening is effective. Screening has in part been responsible for a steady decline in the incidence of CRC since the mid-1980s. Failure to screen may result in CRC presenting with advanced-stage disease. CRC is usually asymptomatic until late in its course when patients may develop lower gastrointestinal bleeding, obstruction, and pain. Randomized clinical trials (RCTs) showed that screening in average-risk individuals (ie, those without a family history of CRC), reduced the incidence and mortality from CRC.[2-10] In contrast to screening programs for lung, prostate, and breast cancer, which focus on early-stage cancer detection, CRC screening facilitates detection of early-stage cancer and precancerous lesions (adenomas), which, when removed, may prevent cancer.[11]

The benefits of CRC screening were recognized 4 decades ago when the American Cancer Society first recommended CRC screening in the late 1970s. The US Preventive Services Task Force (USPSTF) then endorsed screening with fecal occult blood testing (FOBT) and flexible sigmoidoscopy in 1995. This regimen became a funded benefit for Medicare recipients in 1997, colonoscopy was added to the Medicare menu in 2001, and stool DNA testing was added in 2015. Private insurers have largely followed the recommendations of Medicare and provide coverage for some, but not all screening options. Despite the proven benefits of CRC screening and general consensus recommending screening by most professional organizations, only 60% to 65% of individuals older than age 50 years actually undergo one of the recommended CRC screening tests.[12]

One way to improve screening rates is to offer a variety of screening methodologies that would make it easier for patients to pursue screening based on their personal preferences. Consequently, new guidelines released in 2016 from the USPSTF[13] emphasized the benefits of CRC screening in average-risk individuals and listed 8 possible screening programs that could reduce CRC

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mortality. A program is defined as a series of steps, which includes offering a test, performance of test with high quality, and appropriate follow-up of positive and negative tests. Each CRC screening program has a unique potential for mortality reduction, incidence reduction, and possible need for colonoscopy as a part of the screening program and based on the risk level of the individual patient. Decision models used by the USPSTF (MISCAN [Microsimulation Screening Analysis Colorectal Cancer Model], SimCRC [Simulation Model of Colorectal Cancer], CRC-SPIN [Colorectal Cancer Simulated Population Model for Incidence and Natural History]) suggest that screening programs emphasizing fecal testing to detect early-stage cancer result in fewer colonoscopy examinations but are not as effective in reducing the incidence of CRC.[13] For example, a program that starts with annual fecal immunochemical testing (FIT) for patients aged 50 to 75 years is estimated to reduce the incidence of CRC by 47% to 72%. In contrast, a CRC screening program of colonoscopy every 10 years reduces the incidence of CRC by 62% to 88%.[14] However, a colonoscopy program is associated with more than 4000 lifetime colonoscopies per 1000 screening recipients, whereas an annual FIT program results in approximately 1750 colonoscopies per 1000 screening recipients. Complications may occur in 15 of 1000 screening recipients in a colonoscopy program as compared with 10 of 1000 in a FIT program. Some patients may be willing to accept a higher lifetime burden of colonoscopy (with attendant risk) for the potential benefit of never developing CRC and never dealing with colon cancer care (surgery, chemotherapy). Other patients may prefer a less invasive approach, which can effectively reduce mortality, but may be less likely than routine colonoscopy to prevent CRC. These elements should be part of an informed decision-making discussion.

The new recommendations from the USPSTF encourage testing but do not specify a preferred program. **Table 1** summarizes the various CRC screening options and recommendations for CRC screening issued by the USPSTF[13]; the American Cancer Society, American College of Radiology, and the US Multi-Society Task Force on Colorectal Cancer[15]; and the American College of Physicians.[16] Even though having choices may improve screening rates in certain populations,[17] there is little doubt that the variety of options complicates informed decision-making conversations.[18] Each CRC screening option has programmatic steps, and program effectiveness depends on appropriate follow-up of positive and negative tests. All screening programs depend on the performance of high-quality colonoscopy. Elements of high-quality colonoscopy include completeness of the examination, quality of bowel preparation, adequate adenoma detection rates, and appropriate follow-up recommendations.

In the United States, most CRC screening is opportunistic rather than occurring in the context of an organized national or regional program. Exceptions include some health maintenance organizations and the Department of Veterans Affairs, which offer screening to all participants. Whatever program is used, one of the principle determinants of patients' screening behavior is recommendations by their primary care clinicians.[19,20] Messaging to both patients and their primary care clinicians is complicated, and it is not surprising that many patients report that their primary care physician has never discussed CRC screening with them. By increasing the availability of various options for CRC screening, potentially reluctant patients will undergo screening. The aim of this review is to provide tools for health care clinicians and their patients to assist in the process of informed decision making for CRC screening (**Box**).

Box. Supplemental Items

eBox 1. Suggested discussion points for informed-decision making about colorectal cancer screening in average-risk individuals

eTable. Common patient questions about colorectal cancer screening

eBox 2. Quality measures for colorectal cancer screening with fecal blood testing and colonoscopy

Methods

PubMed was searched for publications related to CRC incidence, mortality, and screening. The search was limited to 2008-2016 to update previous searches by the USPSTF[21] and was not intended to be a systematic review because the USPSTF recently completed a new guideline.[13] We focused on new data regarding CRC screening efficacy and program adherence and also on the quality of screening modalities.

Observations

Trends in CRC Incidence and Mortality

In the United States, there is significant regional variation in risk of CRC.[22] The high-risk states in the central Midwest have an incidence of 41.7 to 48.9 cases per 100 000 age-adjusted population, which is much higher than low-risk states (incidence 32.5-37.5 cases per 100 000 age-adjusted population). Women have a lower age-adjusted risk compared with men across every race and ethnicity, but have similar lifetime risk.[1] Black men and women have a higher incidence and mortality compared with men and women of other races and ethnicities in the United States.[1,23]

US CRC mortality declined by 30% between 2000 and 2010.[1] This reduction was observed in all races and in both sexes.[1] These gains might be explained by more effective prevention, earlier detection, and better treatments. Prevention appears to explain some of this reduction because the incidence of new CRC cases has declined over the past decade.[1,24] The reduced incidence of CRC can also be explained, in part, by increased screening rates. If screening identifies patients with precancerous polyps and those polyps are removed, there is strong evidence that the incidence of CRC will decline.[6-9,25,26] In the National Polyp Study, mortality rates were reduced by 53% among individuals who had polypectomy as compared with population-based expected CRC rates from SEER (Surveillance, Epidemiology, and End Results) centers.[11] Because the incidence and mortality for CRC began to decrease before screening was very common, it cannot fullyexplain the effect of screening on these trends.[1]

Other factors account for some of the secular trends of reduced incidence and mortality of CRC. In the past 40 years, tobacco use has decreased, aspirin is more widely used for cardioprotection, over-the-counter nonsteroidal anti-inflammatory drugs are

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Table 1. US Guidelines for Colorectal Cancer Screening in Average-Risk Individuals

	Fecal Occult Blood	Flexible Sigmoidoscopy	Colonoscopy	Multitarget FIT-DNA	Computed Tomographic Colonography
USPSTF[13]					
Recommendation	Yes with highly sensitive gFOBT or FIT	Yes	Yes	Yes	Yes
Interval	Annual	Every 5 years alone Every 10 years with annual FIT	Every 10 years	Every 1 year Every 3 years	Every 5 years
ACS/ACR/MSTF-CRC (2008)[15]					
Recommendation	Yes with highly sensitive test (FIT)	Yes	Yes	Yes	Yes
Interval	Annual	Every 5 years	Every 10 years	Uncertain	Every 5 years
ACP Guidance Statement (2012)[16]					
Recommendation	Yes with gFOBT or FIT	Yes	Yes	Yes	Yes
Interval	Annual	Every 5 years	Every 10 years	Uncertain	Every 5 years

Abbreviations: ACP, American College of Physicians; ACR, American College of Radiology; ACS, American Cancer Society; FIT, fecal immunochemical test; gFOBT, guaiac fecal occult blood testing; MSTF-CRC, US Multi-Society Task Force on Colorectal Cancer; USPSTF, US Preventive Services Task Force.

more available, and hormone less use of replacement therapy by perimenopausal women—all of which may prevent CRC.[27]

Some risk factors are associated with an increased risk of CRC. Age is one of the strongest risk factors for CRC, and in the United States, the population is living longer.[1] High body mass index is a CRC risk factor and the US population has gotten significantly heavier in recent decades.[27] Other western countries have similar risk factor profiles and trends as the United States; yet, these other western countries have not experienced the declining incidence of CRC as has been seen in the United States, which might be explained by more aggressive screening programs as compared with other western countries.[24]

USPSTF CRC Screening Guidelines

In 2016, the USPSTF issued new recommendations for CRC screening.[13] These are summarized in Table 1. The USPSTF recommends that average-risk adults start screening at age 50 years and continue until age 75 years (A recommendation). From age 76 to 85 years, screening decisions should be individualized. Adults who have not been previously screened may be more likely to benefit. Adults who are healthy enough to undergo treatment if CRC is detected and do not have comorbid conditions that would limit life expectancy may benefit (C recommendation). The USPSTF recommended against CRC screening for asymptomatic persons aged 86 years and older.

Identification and Management of Individuals With Higher-Than-Average Risk

The USPSTF screening guidelines for CRC apply only to average-risk individuals. However, a screening strategy is even more important for persons at higher-than-average risk for CRC. It is important to identify individuals with a higher-than-average risk, who require more intensive colorectal evaluation. Answers to 3 questions can help identify these patients: (1) is there a personal history of CRC or adenomatous polyps; (2) is there any history of inflammatory bowel disease, Crohn disease, or ulcerative colitis; and (3) does the patient have a first-degree relative with CRC, and if so, did any relative have CRC at younger than 50 years.

Determining If There Is a Personal History of CRC or Adenomatous Polyps

The first question identifies individuals with prior colon neoplasia who are at risk to develop new colon neoplasia. Recent guidelines provide tools for determining appropriate intervals for colonoscopy surveillance in patients with prior cancer[28] or polyps[29] and are summarized in **Table 2**. Patients with prior CRC or neoplastic polyps have a higher risk of CRC; hence, surveillance with colonoscopy is recommended with interval stratified by risk. In brief, patients with CRC should have a follow-up screening in 1 year, patients with advanced polyps (adenomas >10 mm, with villous histology, or high-grade dysplasia) in 3 years, and patients with low-risk adenomas (1-2 tubular adenomas <10 mm) in 5 to 10 years, depending on the quality of the baseline colonoscopy.[28,29]

Determining If There Is a History of Ulcerative Colitis or Crohn Disease Involving the Colon

The second question identifies individuals with ulcerative colitis or Crohn disease involving the colon. These individuals have an increased risk of CRC, which begins after 8 years of chronic disease.[30] These individuals should be referred for regular surveillance with colonoscopy, even if disease symptoms are under control.

Determining If the Patient Has a First-Degree Relative With CRC and If So, Was Any Relative Diagnosed at Younger Than 50 Years

The third question identifies individuals with familial risk and hereditary syndromes. If there is a family member diagnosed with CRC at younger than 50 years, known hereditary syndromes (eg, familial adenomatous polyposis and Lynch syndrome, which are associated with specific germline mutations) should be considered. Most patients with familial adenomatous polyposis will have family members who developed CRC at younger than 50 years, and nearly 100% will develop CRC during their lifetime. Screening for familial

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Table 2. Colorectal Cancer Screening and Follow-up in Higher-Than-Average–Risk Individuals

Risk Level for CRC	Lifetime Risk of CRC	Initiation of Screening	Follow-up Intervals
Very high–germline mutations			
Familial polyposis	Nearly 100%	During teenaged years	Not applicable; colon resection is recommended
Lynch syndrome	Variable depending on mutation	Age 20-25 years	Colonoscopy every 2 years
High risk			
Ulcerative colitis or Crohn colitis	Increases with time 10%-20%	After 8 years of chronic disease	Colonoscopy every 2 years
Familial risk (first-degree relative ≤60 years)	10%-20%	Age 40 years or 10 years younger than index family member	Colonoscopy every 5 years
Familial risk (first-degree relative >60 years)	10%-15%	Age 40 years or 10 years younger than index family member	Any recommended screening test
High risk (personal history of colorectal neoplasia)			
Personal history of CRC	Uncertain; >10%	1 year after baseline colonoscopy with diagnosis of CRC	Colonoscopy at 3 years then again at 5 years if negative for neoplasia ; colonoscopy at appropriate interval if polyps are detected
High-risk adenoma(s) (adenoma ≥10mm with villous histology, high-grade dysplasia, or >3 adenomas or serrated polyps with dysplasia)	Uncertain; greater-than-average risk		Colonoscopy at 3 years after high-quality baseline colonoscopy with complete removal of polyps
Low-risk adenoma(s) (1-2 tubular adenomas <10 mm)	Uncertain; perhaps similar to average risk		Colonoscopy at 5-10 years after high-quality baseline colonoscopy with complete removal of polyps

Abbreviation: CRC, colorectal cancer.

adenomatous polyposis should be initiated in the late teen years when adenomatous polyps can be detected. New guidelines related to Lynch syndrome were published by the US Multi-Society Task Force on Colorectal Cancer (MSTF-CRC) in 2014.[31] Most of these individuals develop CRC at younger than 50 years, so it is important to initiate screening with colonoscopy well before the expected development of CRC at age 20 to 25 years. If family history indicates risk based on the aformentioned characteristics, referral to experts to consider genetic counseling and appropriate initiation of screening (Table 2) is advised.

Another group of higher-risk patients are those with a first-degree relative with CRC (relative risk [RR] with 1 first-degree-relative, 2.24 [95% CI, 2.06-2.43]; or with >1 first-degree relative (RR, 3.97 [95% CI, 2.60-6.06]).[32] Thus, if average-risk individuals have a 5.5% lifetime risk of CRC, patients with 1 or more first-degree relatives have a lifetime risk or 10% of more. Current recommendations based on expert consensus call for these individuals to initiate screening at age 40 years, or 10 years before the age of the index family member, whichever comes first because there is a higher lifetime risk of CRC. Colonoscopy is generally the preferred form of screening if the index family member had CRC or advanced adenoma before age 60 years.[15]

Interviews about personal and family history should be updated every 1 to 2 years to avoid perpetuation of legacy data obtained during a first, intensive interview, which may not be accurate over time.

Screening Options for Average-Risk Individuals

Guidelines from the USPSTF[13]; the combined guideline of the American Cancer Society, American College of Radiology, Multi-Society Task Force on Colorectal Cancer[15]; and practice recommendations from the American College of Physicians[16] do not favor use of a single specific test (Table 1). A discussion of test performance, programmatic aspects, and downstream consequences can require considerable time, which may be challenging for busy primary care physicians multitasking many other medical issues. Suggested discussion items are summarized in eBox 1 (in the Supplement). Although this education can be provided in printed and electronic media, there is some evidence that discussion of options with a primary care physician is important to achieve high levels of adherence to CRC screening.[19,20] Information to help primary care clinicians facilitate patients' informed decision making are summarized in **Table 3**. Questions commonly asked by patients are summarized in the eTable (in the Supplement).

Stool-Based Tests

Fecal Occult Blood Test and Fecal Immunochemical Test

The primary aim of stool testing for occult blood is detection of early-stage cancer. RCTs in the 1990s found that screening with a guaiac fecal occult blood test (gFOBT) is associated with reduction in CRC mortality (15%-30%) and modest reduction in incidence (17%-20%) compared with unscreened control participants.[2-6] Current guidelines suggest a preference for the sensitive gFOBT (SENSA) or fecal immunochemical test (FIT), which detects human hemoglobin in stool, and testing is recommended every year. The FIT has higher sensitivity compared with older gFOBT for detection of CRC (79% vs 20%-50%) and also for advanced polyps, which may be cancer precursors (20%-50% for FIT vs 11%-20% for gFOBT).[33] There are no RCTs of FIT or the sensitive gFOBT, but it is likely that decreases in CRC mortality and incidence would be greater with FIT

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Table 3. Information for the Primary Care Clinician Screening for Colorectal Cancer

Characteristic	gFOBT[a]	FIT	Multitarget Stool DNA	Flexible Sigmoidoscopy[a]	Colonoscopy	CTC
Evidence source/study design	RCTs of gFOBT	Cohort studies to ascertain sensitivity and specifity	Comparisons of stool DNA, FIT, and colonoscopy	RCTs in 4 countries	Case-control	Comparisons of CTC and colonoscopy
One-time sensitivity of test for CRC	20%-50% (RCTs)	79% (no RCTs)	≤92% (no RCTs)	95% In portion examined; 75% overall	95%	Unknown (likely >90%)
Sensitivity of test of precancerous adenomatous (polyps ≥10 mm)	11%-20% (RCTs)	20%-50% (no RCTs)	42% (no RCTs)	90% In portion examined	90%	90%
CRC mortality reduction	15%-30%	Uncertain (but likely better than gFOBT because of higher CRC sensitivity)	Unknown	22%-31% (RCTs)	45%-77% (no RCTs)	Unknown
CRC incidence reduction	20% Annual screening; 17% biennial screening	Uncertain (but likely better than gFOBT because higher adenoma detection)	Unknown	18%-23% (RCTs)	31%-88% (no RCTs)	Unknown
Attributes of test	Detects blood in stool sample; requires 3 samples; requires special diet	Detects human hemoglobin in stool sample; requires 1 sample; no special diet required	Detects specific CRC-related mutations and human hemoglobin in stool sample combined with FIT; requires 1 sample; no special diet required	Flexible endoscope inserted into the lower colon 55-60 cm	Flexible endoscope inserted into the entire colon; usually performed with sedation	Provides CT imaging; detects lesions >5 mm; smaller polyps are generally not reported
Patient requirements	Home-based test; 3 stool samples are smeared on test card; test card is mailed or brought to laboratory for analysis	Home-based test; 1 stool sample is smeared on test card; test card is mailed or brought to laboratory for analysis	Home-based test; 1 stool sample is obtained with kit and mailed to reference laboratory for analysis	Bowel preparation; office visit; can be performed without sedation with minimal loss of work time	Bowel preparation; performed at endoscopy center or hospital endoscopy unit; requires sedation, designated driver, and missed work day	Bowel preparation; visit to imaging center
Interpretation of positive test results	Compared with patients with negative test results, risk of CRC has 3- to 4-fold increased risk; advanced adenoma (polyps ≥10 mm) has 2-fold increased risk	Compared with patients with negative test results, risk of CRC and polyps is greater	Compared with patients with negative test results, risk of CRC and polyps is greater	If polyps are seen, risk of additional colon polyps in the proximal colon is greater also	Polyps, if seen, can usually be removed during the procedure; if CRC is detected, tissue diagnosis can be confirmed and patient is referred to surgery	Compared with patients with negative test results, risk of CRC or polyps is greater
Next steps after positive test results	Colonoscopy is recommended because of increased risk of CRC and advanced polyps; colon surveillance is recommended if CRC or polyps are detected	Colonoscopy is recommended because of increased risk of CRC and advanced polyps; colon surveillance is recommended if CRC or polyps are detected	Colonoscopy is recommended; colon surveillance is recommended if CRC or polyps are detected	Colonoscopy is recommended because of increased risk of CRC and advanced polyps; a second bowel preparation is required; colon surveillance is required if polyps or CRC are detected	Colon surveillance is recommended if CRC or polyps are detected	Colonoscopy is recommended; a second bowel preparation is required (unless CTC can be performed on the same day); colon surveillance is required if CRC or polyps are detected
Interpretation of negative results after a single test	CRC or advanced polyps risk remains; 50%-80% of patients with CRC may have a negative test	CRC or advanced polyps risk remains; 14%-31% of patients with CRC may have a negative test	CRC or advanced polyps risk remains; 8% of patients with CRC may have a negative test	CRC or advanced polyps risk in the portion of colon not examined remains	Likelihood of CRC diagnosis in ≤10 years is low	Risk of having CRC or undetected advanced polyps is small
Next steps after negative test results	Annual programmatic testing; further testing can detect CRC not detected in an earlier round	Annual programmatic testing; further testing can detect CRC not detected in an earlier round	Uncertain interval for repeat testing; USPSTF recommends either a 1- or 3-year interval	If combined with FIT, repeat testing in 10 years; if not combined with FIT, repeat testing in 5 years	USPSTF recommends repeat testing in 10 years	USPSTF recommends repeat testing at a 5-year interval

(continued)

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Table 3. Information for the Primary Care Clinician Screening for Colorectal Cancer (continued)

Characteristic	gFOBT[a]	FIT	Multitarget Stool DNA	Flexible Sigmoidoscopy[a]	Colonoscopy	CTC
Harms	No direct harm; false reassurance of negative test may result in lower adherence to subsequent rounds of testing	No direct harm; false reassurance of negative test may result in lower adherence to subsequent rounds of testing	No direct harm; false reassurance of negative test may result in lower adherence to subsequent rounds of testing	Very low risk of perforation (1/10 000); patient discomfort	Bleeding risk (3/1000); perforation risk (1/1000); cardiovascular events risk (1/1000)	Very small radiation exposure; minor discomfort
Quality issues	Adherence to annual repeat testing if negative; adherence to colonoscopy if positive	Adherence to annual repeat testing if negative; adherence to colonoscopy if positive	Unknown	High-quality inspection required	High-quality inspection with complete examination to cecum; adequate preparation; adequate adenoma detection rate; complete removal of polyps	Requires good bowel preparation; interobserver reader variability
Costs	Low initial cost Downstream costs include repeat testing annually if test is negative, cost of ensuring adherence to repeat testing if test is negative, colonoscopy if test is positive, colon surveillance if polyps or CRC are detected	Low initial cost Downstream costs include repeat testing annually if test is negative, cost of ensuring adherence to repeat testing if test is negative, colonoscopy if test is positive, colon surveillance if polyps or CRC are detected	High initial cost ($500-$600) Downstream costs include colonoscopy if test is positive, colon surveillance if polyps or CRC are detected, repeat testing in 1-3 years if test is negative	Low initial cost Downstream costs include repeat testing if test is negative, cost of ensuring adherence to repeat testing if test is negative, colonoscopy if test is positive, colon surveillance if polyps or CRC are detected	High initial cost ($500-$3000) and can be higher with anesthesia-administered sedation Downstream costs include infrequent repeat testing if test is negative (at 10 years), colon surveillance if polyps or CRC are detected, cost of harms, colon surveillance if polyps or CRC are detected	High initial cost ($500-$600) Downstream costs include colonoscopy if test is positive, colon surveillance if polyps or CRC are detected, repeat testing if test is negative
Cost-effectiveness models	Cost effective relative to other medical interventions; most models show similar lifetime costs compared with other CRC screening tests	Cost effective relative to other medical interventions; most models show similar lifetime costs compared with other CRC screening tests	Unknown	Cost effective relative to other medical interventions; most models show similar lifetime costs compared with other CRC screening tests	Cost effective relative to other medical interventions; most models show similar lifetime costs compared with other CRC screening tests	More costly than other screening programs depending on rate of referral for colonoscopy

Abbreviations: CRC, colorectal cancer; CTC, computed tomographic colonography; FIT, fecal immunochemical test; gFOBT, guaiac fecal occult blood test; RCT, randomized clinical trial; USPSTF, US Preventive Services Task Force.

[a] RCTs demonstrated reduction in CRC mortality for the gFOBT but not for the sensitive gFOBT or FIT.

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than for gFOBT because of the more favorable operating characteristics of the FIT. There is evidence that patient adherence to the screening program may be better with FIT than for gFOBT since FIT requires testing of only 1 stool sample and no dietary restrictions, whereas gFOBT requires 3 stool samples and dietary restrictions.[34]

Harms of Fecal Blood Testing

Not all patients with CRC will have a positive test result, which could delay CRC diagnosis. False reassurance after a negative screening test is a potential harm if it results in poor adherence for subsequent testing. The FIT program is especially vulnerable to this harm, which highlights the importance of informed decision making so that patients fully understand that a negative test does not eliminate the possibility of having CRC and needing repeat testing annually.

Key Elements of Informed Decision Making and Quality of the Fecal Blood Test Screening Program

The effectiveness of FIT and gFOBT depends on adherence to several steps of the stool-based CRC screening program: (1) completion of the initial stool tests; (2) undergoing colonoscopy if the test is positive; and (3) having repeat stool testing every year if the test is negative. Rates of patient adherence to a first round of testing are 60% to 80%.[35-40] Studies[34-36] have shown that some people are substantially more likely to complete a stool blood testing program than one exclusively based on periodic colonoscopy.

Following a positive fecal test for occult blood, approximately 70% to 90% of patients undergo follow-up colonoscopy.[5,38] Screening will not be effective for those patients not undergoing follow-up colonoscopy because early-stage CRC or polyps are not detected. Patients should fully understand the importance of adherence to the full screening program for it to be effective. Treating physicians should monitor patients' adherence to the full CRC screening program as a quality measure.

The rationale for repeat stool testing for occult blood or DNA if the initial testing is negative relates to false-negative tests (21% with FIT). Patients who have polyps or cancers that test negative for fecal occult blood or DNA may have a positive subsequent test and still have earlier CRC detection than if they were not tested at all.[2-4,40] Few studies have examined how many patients undergo repeat testing following negative stool tests. Repeat testing rates range from less than 30%[5,39] to as much as 75%.[40] High repeat testing rates require extensive infrastructure to generate and follow up on patient reminders. Adherence to repeat testing after an initial negative test result should be monitored as a quality indicator if this program is used.

Multitarget FIT and Stool DNA Testing

A multitargeted FIT and stool DNA test is newly approved by the US Food and Drug Administration (FDA) and the Centers for Medicare & Medicaid Services (CMS) for CRC screening. This test can detect specific mutations in stool samples that are associated with colon neoplasia, and also it incorporates a FIT test for occult blood. In a recent study, patients had stool DNA testing and FIT before undergoing colonoscopy. The sensitivity of stool DNA testing plus FIT was superior to FIT alone for CRC (92% vs 74%) and for advanced polyps.[41] The false-positive rate of 14% is higher than FIT (5.1%) and would result in higher rates of unnecessary colonoscopy. The effect of FIT and DNA stool screening on the incidence and mortality of CRC is not known. The 3-year testing interval suggested by the manufacturer is based largely on the premise that the test is more sensitive than FIT alone and could be performed less often. The USPSTF models suggest that performing the test every year could result in higher rates of reduced CRC mortality; the USPSTF recommendation of a 1- or 3-year interval is based on the models.[13]

Harms of Multitargeted FIT and Stool DNA Testing

The multargeted FIT and stool DNA test may fail to identify some patients with CRC and provide false reassurance resulting in poor adherence with repeat testing. There are no data on serial testing available. If the test result is positive, and a subsequent colonoscopy is negative, there may be uncertainty about whether the stool DNA was truly a false-positive test. This issue has been raised because if the DNA portion of the test is positive, there remains the possibility that a neoplastic lesion may have been present but not detected at colonoscopy. At the time of this publication, there is no recommendation for more intensive follow-up because it is not possible to determine which component of the test was positive—either the stool DNA or the FIT.

Key Elements of Informed Decision Making and Quality of the Multitarget FIT and Stool DNA Test

Patients should fully understand and be prepared to follow all the steps required for CRC screening programs based on stool testing, which include a need to undergo colonoscopy if the stool test is positive. The cost-effectiveness is uncertain, and it will depend on the cost of the various tests and the rate of referral for colonoscopy. The recommended interval of 1 year or 3 years is uncertain and not based on evidence. There are currently no studies on repeat testing.

This is a new test for which quality measures have not yet been fully developed. Quality metrics to be considered are rate of test completion, rate of follow-up colonoscopy for positive test results, and quality control in the laboratory performing the test to ensure reproducibility of results.

Structural Examinations of the Colon

Computed Tomographic Colonography

Computed tomographic colonography (CTC) provides 2- and 3-dimensional imaging of the colon. The test requires a bowel preparation and uses very low-dose radiation. CTC has a similar polyp detection rate as colonoscopy when performed and interpreted by experts (polyps ≥10 mm [90%] and for polyps 6-9 mm [70%-80%]).[42] There are no studies demonstrating CRC incidence and mortality reduction, but decision models project significant reductions based on the sensitivity of the test.[14,43,44] Lesions smaller than than 6 mm are generally not reported; therefore, patients may have undetected small polyps. The ability of CTC to detect flat serrated colon lesions is unknown, which may be an important limitation since there is increasing evidence that flat serrated polyps represent an important pathway to CRC.[45] Since the serrated polyp pathway may account for more than 20% of CRCs, further research is needed to determine if CTC can detect these lesions and, even if not, how well it reduces CRC mortality relative to other screening modalities.

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Key Elements of Informed Decision Making and Quality for CTC

Potential benefits of choosing CTC as a screening test include its ability to visualize the entire colon. It is a safe option, and complications are rare. Patients should understand the complete program that is required for CTC to be an effective CRC screening modality. The procedure requires a complete bowel preparation. If the test is positive for polyps, a colonoscopy will be recommended, which will require a second bowel preparation unless it is possible to perform same-day colonoscopy. Patients should understand that they may have small polyps (<6 mm) that will not be reported. They should be aware that imaging can identify other abdominal pathology that can be of uncertain clinical significance resulting in unnecessary testings. Unexpected findings occur in 25% to 80% of average-risk participants undergoing a CTC and most are incidental and clinically unimportant. Algorithms developed by the American College of Radiology have been proposed that identify the subset of individuals (5%-15% of all screened for CRC) with significant extra-intestinal lesion(s), such as abnormalities of adrenal gland, kidney, pancreas, and lymph nodes, which will need some other evaluation.[46] Because of the need for multiple procedures and follow-up testing associated with incidental findings, the costs associated with CTC are high.[13] The total cost for CTC screening depends largely on the cost of the initial CT imaging, rate of referral for colonoscopy (estimates range from 10%-25%), and rates of evaluation for extra-intestinal findings. Patients should understand that each study is associated with a small amount of radiation exposure.[47]

Flexible Sigmoidoscopy

Flexible sigmoidoscopy examines the most distal one-third of the colon. It is generally performed without sedation in the United States and elsewhere. The test requires a less-complete bowel prepararion than colonoscopy. RCTs showed that flexible sigmoidoscopy reduces CRC mortality (22%-31%) and incidence (18%-23%) by detection of early-stage cancer and detection and removal of cancer precursor lesions.[7-10] Flexible sigmoidoscopy can be performed by primary care physicians or by nonphysicians (nurse, nurse practitioner, physician assistant). Observational studies suggest that colonoscopy may be more effective than flexible sigmoidoscopy, although the benefit of colonoscopy in the proximal colon may be less profound than in the distal colon.[48] Despite the RCT evidence, flexible sigmoidoscopy is seldom used for CRC screening in the United States. There are several factors leading to its low rate of use. If patients have polyps detected by flexible sigmoidoscopy, a full colonoscopy with a complete bowel preparation will still be required. Patient preferences for sedation may also contribute to low use of flexible sigmoidoscopy in the United States. In addition, with increasing age, there is an increased risk of proximal CRC and polyps, which are less likely to be detected with flexible sigmoidoscopy.[48] Flexible sigmoidoscopy is not recommended for patients with a high risk for CRC, including those with a history of prior colon neoplasia, family history of CRC, or hereditary CRC.

The 2016 USPSTF guideline recommends an flexible sigmoidoscopy every 10 years, combined with annual FIT or sensitive gFOBT every 5 years if not combined with a fecal blood test.[13] This recommendation is based largely on 1 study[10] showing some benefit of combined screening compared with flexible sigmoidoscopy alone.

Harms of Flexible Sigmoidoscopy

Flexible sigmoidoscopy is safe, and serious complications are rare, with bowel perforation occurring in fewer than 1 in 10 000 cases. Because the procedure is often performed without sedation, there is mild-to-moderate patient discomfort.

Key Elements of Informed Decision Making and Quality

Benefits of flexible sigmoidoscopy for screening include high safety and low cost. Patients should be informed that flexible sigmoidoscopy may not detect pathology in the proximal colon. If polyps are found, colonoscopy would be recommended to evaluate the proximal colon.

Colonoscopy

Although there are no RCTs of screening with colonoscopy demonstrating a reduction in mortality, there is evidence that detection and removal of precancerous polyps reduces the incidence and mortality of CRC.[11,25,26] The potential advantages of colonoscopy over flexible sigmoidoscopy include the ability to visualize the proximal colon, remove polyps during 1 examination, and perform examinations at 10-year intervals. Based on decision models used by the USPSTF, colonoscopy is likely to lead to the lowest CRC incidence than is achievable by any other screening programs, which may be due to polyp detection and removal.[11,14] There are concerns that colonoscopy may be less effective in lesion detection in the proximal colon compared with the distal colon,[48] although recent studies demonstrate reductions in proximal colon cancer after colonoscopy.[49-51] If important proximal lesions are not detected, the potential benefit of colonoscopy over flexible sigmoidoscopy would be diminished, and there would be a higher risk of harms compared with other forms of screening.[13] Currently, trials in the United States and Spain[35] are comparing FIT and colonoscopy with end points of CRC mortality and incidence; the NordICC (Northern-European Initiative on Colorectal Cancer) trial is comparing colonoscopy vs no offer of screening in 4 countries (Norway, Sweden, Poland, Netherlands).[52]

Harms of Colonoscopy

Aside from colonoscopy alone being an effective CRC screening test, all screening programs include colonoscopy if the initial test suggests the presence of cancerous lesions. Colonoscopy is an invasive procedure, usually performed with conscious or deep sedation and there may be serious risks.[13] In a large prospective study, the risk of important adverse events such as serious bleeding, bowel perforation, or cardiopulmonary events leading to hospitalization within 30 days of the procedure was 3 to 4 per 1000 procedures.[53] In patients older than 65 years, the risk of serious gastrointestinal events within 30 days of colonoscopy (bleeding, perforation) was 7 per 1000 procedures,[54] and 16.4 per 1000 had unplanned hospital visits within 7 days of the procedure.[55] The risk for complications is increased with age, comorbid conditions, polypectomy, and use of anticoagulation drugs.[53,55] These potential harms should be carefully weighed in the evaluation of risk and benefit for screening and surveillance.[29]

Key Elements of Informed Decision Making and Quality of Colonoscopy

Colonoscopy is central to all screening programs. The quality of the colonoscopy, including complete inspection, detection of impor-

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tant polyps, and complete lesion removal is an important aspect of the effectiveness of colonoscopy for CRC screening (eBox 2 in the Supplement). The following quality metrics have been defined[56,57] and should be monitored as part of a high-quality practice: documentation of adequate bowel preparation; the ability to intubate the cecum; and adenoma detection rates. A key drawback of colonoscopy includes its relatively high expense.

Other Screening Issues

When to Initiate Average-Risk Screening

Currently, most guidelines in the United States recommend the initiation of screening at age 50 years for both men and women (Table 1). When compared with 50- to 54-year-old white men, the risk of CRC and advanced polyps is approximately 50% lower in 50- to 54-year-old women who are not black.[58] Women have reduced risk for CRC until menopause, likely due to hormonal protection, after which risk rises at a similar rate to men.[1,59,60] If a risk-based approach to screening is adopted in the future, women who are not black may be able to initiate screening at a later age. A risk-based approach does not diminish the importance of CRC screening in women who have a similar lifetime risk as men—it simply recognizes that average-risk women who are not black develop CRC later in life than men.

Some practice recommendations call for early screening of all black men and women starting at 40 years[16] or 45 years.[61] In the United States, black individuals have a higher incidence and mortality compared with other races.[23] It is not clear if early screening at younger than 50 years is preferable to increasing efforts to initiate screening at 50 years among black men and women.

When to Stop Average-Risk Screening

The USPSTF recommends that screening stop after age 85 and should be individualized from age 76 to 85 years, depending on results of prior screening.[13] This recommendation is based on decision modeling, which suggests that benefits of screening (extended life as a result of prevention or early detection of CRC) may diminish if the individual is unlikely to live more than 7 years due to comorbid conditions.[13] There is evidence that risk of adverse events associated with colonoscopy increase among adults older than 65 years.[54,55] Therefore, at some point in every life, risk will outweigh benefit. For older individuals, informed decision making should account for comorbid conditions and the likelihood of benefit, which is unlikely if life expectancy is limited by other medical conditions.

If Screening Colonoscopy Is Negative for Neoplasia, Determining Whether Interim Fecal Blood Testing Should Be Performed Before the Next Scheduled Colonoscopy

There is little evidence that interval testing after a normal colonoscopy is of benefit. Recent studies show that the risk of finding large polyps is low when compared with an initial negative screening colonoscopy.[62] Therefore, this practice is discouraged.[15]

Management of Antithrombotic Agents for Patients Undergoing Colonoscopy

Every screening program can ultimately lead to the performance of colonoscopy if initial noncolonoscopy screening tests are positive.

Table 4. Recommendations for Management of Antithrombotic Agents in Patients Undergoing Elective Colorectal Cancer Screening

Antithrombotic Drug Categories	Recommendation for Elective Colonscopy
Antiplatelet Drugs	
Monotherapy with aspirin (acetylsalicylic acid) or nonsteroidal anti-inflammatory drugs	Continue standard dose
Monotherapy with thienopyridine	Discontinue for 5 days before switch to aspirin
Dual therapy with aspirin plus thienopyridine	
New drug-eluting stent	Defer elective colonoscopy for 12 moonths; afterwards stop thienopyridine at least 5 days before procedure
New bare metal intracoronary stent	Defer elective procedure for at least 30 days; afterwards stop thienopyridine at least 5 days before procedure
New acute coronary syndrome	Defer elective procedure for at least 90 days; afterwards stop thienopyridine at least 5 days before procedure
Long-term dual therapy	Stop thienopyridine at least 5 days before procedure; continue aspirin
Anticoagulant drugs	
Short-term anticoagulation	Defer elective procedures until completion of course of anticoagulation
Long-term anticoagulation, thrombotic risk assessment	
Low risk	Discontinue anticoagulation for appropriate drug-specific interval; afterwards restart on day of procedure assuming adequate hemostasis
High risk	Bridge therapy using shorter acting anticoagulant which can be stopped prior to procedure

Adapted from Acosta et al.[63]

Many patients older than 50 years are taking antithrombotic agents for cardioprotection. Since polypectomy should be anticipated at any colonoscopy examination, precautions should be taken to reduce the risk of postpolypectomy bleeding without compromising cardiovascular health during the periprocedure period. The recommendations for management fall into 2 major categories—antiplatelet drugs and anticoagulants—and were recently updated[63] and summarized in **Table 4**.

Conclusion

The 2 most commonly used CRC screening programs in the United States and throughout the world are fecal blood test and colonoscopy. It remains unclear which program is most effective. Ultimately a patient's willingness to adhere to the complete CRC screening program and quality of that program determines its overall effectiveness.

The fecal blood test and colonoscopy programs take very different approaches to screening. FIT is noninvasive multistep program. Mortality reduction is primarily achieved by detecting CRC at an early curable stage. There is also modest reduction in incidence, which contributes to mortality reduction because of detection and removal of polyps.[5,6] Programmatic aspects of this test are critical

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to effectiveness. Positive tests should be followed by colonoscopy; a negative test should be followed by repeat annual testing. Patients should understand that failure to adhere to the programmatic aspects of this test will render the program less effective. Quality metrics should be monitored in a high-quality practice (eBox 2 in the Supplement).

Colonoscopy can accurately detect early-stage CRC and polyps. The detection and removal of polyps has been shown to reduce the incidence and mortality of CRC.[7-11,25,26] Colonoscopy is invasive and associated with some risk. Two large ongoing studies are comparing programmatic FIT and colonoscopy; a third study compares colonoscopy with no screening. Colonoscopy is central to every screening program described in this review. The performance of high-quality colonoscopy (eBox 2 in the Supplement) is essential for effectiveness of every CRC screening program.[56,57]

The National Colorectal Cancer Roundtable is a consortium of primary care clinicians, specialists, and public organizations committed to the reduction of CRC. This group set a goal of screening 80% of the average-risk population older than 50 years by 2018. Uptake of screening and adherence to a screening program over time are key determinants of the effectiveness of CRC screening—a process that begins by engaging patients in informed decision making. Key patient questions are summarized in the eTable in the Supplement and could provide a reference for patients to assist with this process.

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